

Received 23 July 2025, accepted 22 August 2025, date of publication 26 August 2025, date of current version 3 September 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3603088

RESEARCH ARTICLE

Advanced Encryption Using Generative Adversarial Network for Enhancing Security of Non-Fungible Tokens (NFTs)

ARNAV GUJARATHI¹, PRATHAM OZA¹, AND ASISH BERA¹, (Senior Member, IEEE)

Department of Computer Science and Information Systems, Birla Institute of Technology and Science, Pilani, Rajasthan 333031, India

Corresponding authors: Asish Bera (asish.bera@pilani.bits-pilani.ac.in) and Arnav Gujarathi (f20200066p@alumni.bits-pilani.ac.in)

This work was supported in part by the New Faculty Seed Grant (NFSG) and Cross-Disciplinary Research Framework (CDRF) under Grant C1/23/168 and in part by the Birla Institute of Technology and Science (BITS) Pilani, Pilani Campus, Rajasthan, India.

ABSTRACT Non-fungible tokens (NFTs) represent a unique form of digital asset stored on a blockchain, encompassing a wide array of assets from digital art to real-world commodities. While NFTs offer robust security through decentralization and smart contract enforcement, they are not impervious to cyber threats. Traditionally, NFTs are transferred over blockchain platforms via smart contracts. The primary objective of this study is to introduce a cryptographic framework that integrates an advanced encryption algorithm layer to fortify NFT image transfers, reinforcing content protection in an evolving digital ecosystem. Various pixel-based encryption algorithms have been implemented using image processing techniques and compared considering their levels of encryption and execution times. Based on the comparison, an advanced encryption algorithm has been devised with an added level of encryption and fast execution time. The proposed encryption algorithm leverages alpha composition with a randomly generated image obtained using a Generative Adversarial Network (GAN). The GAN-generated image is infused with Gaussian noise to deter decryption by unauthorized algorithms, thereby enhancing resilience against cryptanalysis. The encrypted image metadata would be transferred through both on-chain (i.e., blockchain) and off-chain methods to facilitate efficiency and security during NFT transfers. The image decryption protocol mandates the new owner/receiver of NFT to possess the correct private identification, transaction key, and off-chain data, ensuring exclusive access while automatically revoking ownership from the previous holder upon successful transfer. This approach not only secures the transfer process but also minimizes the risk of data leaks, as intermediaries should not have access to the complete metadata. This cryptographic approach not only safeguards digital assets but also aligns with the futuristic vision of secure transactions in the metaverse and Web 3.0. The experimental analysis based on the security level and encryption time based on multiple encryption algorithms justifies the benefits of the proposed method.

INDEX TERMS Applied cryptography, alpha composition, blockchain, content protection, encryption, decryption, Generative Adversarial Network (GAN), non-fungible token (NFT).

I. INTRODUCTION

Web 3.0 is a decentralized internet built on blockchain, enabling secure, peer-to-peer digital ownership. The non-fungible tokens (NFTs) play a key role by allowing users to own, trade, and authenticate digital assets without

intermediaries. NFTs represent a transferrable right to digital assets stored on a blockchain [1]. These digital assets encompass a wide variety of real-world commodities such as art, gaming items, collectibles, music, supply chain, social media, and more [2], [3]; which altogether imply metaverse concept into a new direction [4]. NFT is a type of cryptocurrency derived from smart contracts on Ethereum [5]. They differ from classical cryptocurrencies

The associate editor coordinating the review of this manuscript and approving it for publication was Jeon Gwanggil¹.

like Nakamoto and Bitcoin [6], where all the coins are equivalent and indistinguishable. NFTs lack fungibility and, being unique, cannot be traded on a like-for-like basis. This attribute makes them ideal for singularly identifying something or someone. In terms of security, NFTs have several benefits compared to traditional digital assets. NFTs use Ethereum blockchain technology, a decentralized platform for creation and execution of smart contracts and decentralized applications [7]. NFTs have been present in the digital asset landscape since 2014. However, NFTs have received mainstream attention since early 2021 and are a popular application of FinTech and the cryptocurrency ecosystem. Moreover, artificial intelligence (AI) models are integrated with blockchain and NFTs to manage ownership and trustworthiness for data exchange in a transparent and secure way [8]. Popular NFTs include Beeple's digital art sale for \$69 million, Jack Dorsey's first tweet auctioned for \$2.9 million, and the famous Bored Ape Yacht Club collection, among others. Also, the security aspects of cyber-physical systems (CPS) for smart cities using NFT's have been studied [9]. The Aegis has been developed as a privacy-preserving protocol for allowing the users to maintain their privacy of NFT ownership status [10]. Ethereum Virtual Machine (EVM) employs the network of participant computers for carrying out these transactions and in turn reward participant computers through minting of NFTs. NFTs secure asset usage rights and manage their distribution and ownership in the public records while keeping an immutable history of the asset's ownership information. The NFTs (e.g., images) are usually stored in a decentralized storage network or on an off-chain network. The transfer of NFTs involves two transactions, first being the transfer of NFT metadata from seller's digital wallet to buyer's digital wallet. The asset metadata, along with its ownership metadata, is uploaded and persistently saved on the blockchain. This transaction deducts the price of NFT asset from the buyer's digital wallet. The second transaction is creation of a verifiable credential for an end-user to complete the transfer of ownership of NFT asset.

Although NFTs have a tremendous potential impact on the present decentralized markets and future business opportunities, the associated technology is in early stages. While NFTs do offer robust security through decentralization and smart contract enforcement, they are not impervious to cyber threats. Particularly, image NFTs have high economic values and are vulnerable to tampering. Thus, there is a high risk of illegal attempts to compromise the authenticity during transfer. Some popular cases of stealing NFTs have been discussed ahead.

One of the most infamous NFT art thefts occurred in 2021, when a hacker exploited a vulnerability in an NFT marketplace to steal an artwork by the renowned digital artist Beeple [11]. The hacker gained unauthorized access to the platform's smart contract and transferred the valuable NFT to their own wallet, effectively stealing the artwork and its

associated value. In 2016, the Decentralized Autonomous Organization (DAO) hack shocked the cryptocurrency community, resulting in millions of dollars-worth of Ether being stolen due to a smart contract vulnerability [12]. These hacks showcase potential risks associated with smart contract vulnerabilities. Though NFTs offer promising opportunities, it is crucial to prioritize content protection while educating users for seamless integration into a digital ecosystem [13].

Image processing techniques play a significant role in safeguarding digital media during transmission by providing a more robust defense against evolving cyber threats. It plays a crucial role in security applications, aiming to protect images from tampering and to enhance resistance against unauthorized retrieval or automated identification by bots. This significance is notably evident in CAPTCHA (Completely Automated Public Turing test to tell Computers and Humans Apart) security and extends to digital media transfer [14], [15], where additional security measures are essential to thwart hacking, phishing, and other potential compromises [16]. Encrypting images with randomly generated hash functions based on RGB values can significantly reduce the risk of unauthorized access.

At first, this research analyzes the security of present image NFT transfer methods. Then, image processing techniques have been applied to enhance the security level by adding an encryption layer to the input image NFT. A key limitation of existing transfer systems is that the metadata of the original NFT image is uploaded during on-chain transfers, rendering the transferred NFT susceptible to third-party attacks. This vulnerability, however, can be mitigated by transmitting an encrypted image in place of the actual NFT metadata.

In this work, multiple pixel-based encryption algorithms have been implemented and their performances have been compared based on the level of encryption and required execution time [17]. Employing a randomizer on a collection of hash functions may exponentially increase computational possibilities, but it also heightens vulnerability to AI-based decryption attempts [18]. Traditional encryption methods, involving hashing with system-generated transaction and private keys, ensure data privacy between the sender and receiver [19]. However, hashing method is time and resource intensive, and thus not suitable for real-time transactions. Hence, faster and efficient encryption and decryption techniques are crucial.

In addition, another pixel-level technique has been developed for on-chain transfer by combining two images based on RGB values using a 2:1 multiplexer (MUX), with the remaining data sent via off-chain channels. The blockchain will send the encrypted image metadata. No intermediaries could have access to entire metadata at all times, thus enhancing security (Sec IV). However, the problem of this method is the potential decryption functions used in select lines of the MUX. To overcome this challenge, the proposed method utilizes the alpha-channel blending for encryption.

The alpha channel of a visual image (RGB) represents the opacity of the color, denoted as RGBA. Here, the alpha composition technique based on the RGBA values has been used for encryption. Multi-layer alpha blending with hardware accelerations and optimizations could further decrease pixel-level access time from linear to constant, thereby enhances the efficiency [20].

In the proposed algorithm, Generative Adversarial Networks (GANs) are used for generation of unique random images. These random images are used as input along with original image NFT in alpha composition. A GAN-based image-to-image translation allows modification of images in a very realistic manner, thus posing threat to authenticity of original image content [21]. The generated images with minor differences can provide an advantage of further masking the random image used in alpha composition while in transit. This technique might mislead attackers into believing that they have decrypted the original image NFT. Whereas, in reality, they have used a wrong image for the alpha-blending decryption process. Moreover, the Gaussian noise has been added to avoid easy detection using machine learning (ML) tools and decryption algorithms [22]. Artificially generated noise can catalyze a fundamental shift towards more privacy-conscious and effective image encryption. The encrypted image will be shared on the blockchain in SHA-256 (Secure Hash Algorithm 256-bit) format. The necessary decryption information that would be sent off-chain includes image metadata used in alpha composition, alpha values used, and transaction ID. In this scenario, neither the users nor the system has enough data to reconstruct the entire image NFT at any point of time. For off-chain data sharing, conventional OTP systems or QR code generation methods can be used, as those are harder to decrypt and are refreshed in every 15–30 seconds to prevent any possibility of data breach.

The encrypted metadata can be targeted by hackers when it is in transit on either off/on-chain. Thus, it is essential to minimize time lapse between the transfer through secure wallets, like MetaMask [23]. The aforesaid encryption technique could serve as an additional secure layer in the present NFT transfer process. Future scope of this may even be to check the authenticity of such image NFTs or creating extensions for browsers (e.g., Google Chrome) which work in sync with these platforms to add another security layer for transactions of image NFTs. Overall, the proposed scheme will deter any third party to access the original image completely. We aim to contribute to the ongoing discourse in the future of safe and secure media transfer in the digital world, one characterized by trust, empowerment, and mutual benefit for all stakeholders.

This paper aims to enhance the image NFT transfer security to avert theft cases [24] (Section II-B). The proposed algorithm emphasizes on sending encrypted image metadata via on-chain (blockchain) and off-chain routes to reduce the vulnerability to external access. As some data is sent off-chain, entire image NFT metadata is not available at once

to either the users or the transfer system for illicit access. Techniques like Multi-Factor Authentication (MFA) [25] may also be included in future improvement of this technique to enhance security in the off-chain metadata transfer. With increased security, the NFTs may see a further boost in their usage and find newer applications in the industry. In summary, the contributions of this paper are:

- Analyzing the NFT transfer system for image NFTs to identify and mitigate security gaps.
- Compare various encryption algorithms based on computation time and encryption levels to devise a novel algorithm to enhance image NFT transfer security.

The paper is organized as follows: Section II summarizes related works on NFTs and image processing techniques. Section III describes the proposed methodology. The experimental results are discussed in Section IV, and the design choices and its advantages are presented in Section IV-A. Possible attack schemes are summarized in Section IV-B, followed by a robustness analysis in Section IV-C, and the conclusion given in Section V.

II. BACKGROUND WORKS

A. NON-FUNGIBLE TOKENS

Non-Fungible Token (NFT) [26] is a type of cryptocurrency that is derived by the smart contracts of Ethereum [27]. NFT was firstly proposed in Ethereum Improvement Proposals (EIP)-721 [28], and further developed in EIP-1155 [29]. By using the smart contracts, creators can prove the existence and ownership of digital assets in the form of videos, images, event tickets, etc. [30]. The creators can also earn royalties from every successful trade on any NFT market or by peer-to-peer exchanging. Full-history tradability and deep liquidity enable NFT to become a promising intellectual property (IP) protection solution. Early investors obtained thousand-fold returns by selling unique digital collectibles. CryptoPunks [31], among the first NFTs on Ethereum, created more than 10000 collectible punks and promoted the ERC-721 standard. CryptoKitties [32] hit the market in 2017 where participants fiercely competed at high prices to auction the rare cats, with the highest bid being more than 999 ETH3 (equally 3M USD). Thousands of fans from around worldwide have collected over 7.6 million top shot moments, building the roster of rookies, vets, and rising star players [33].

NFTs represent unique, indivisible digital assets, often linked to digital or physical entities. The centralization of NFT marketplaces may pose security risks, including the possibility of platform breaches, data leaks, and unauthorized access to users' digital assets. Hackers may attempt to gain unauthorized access to NFT wallets or marketplaces to steal the private keys or credentials of users. With access, they can transfer or sell the NFTs associated with that account. Dynamic NFTs expand the concept of digital ownership by introducing programmability, adaptability, and interactivity. A blockchain-based architecture has been developed for

expanding NFTs towards the expansion of IoT-enabled smart assets [9]. It enables NFTs to employ fog computing for all users and smart devices connected in a cyber-physical system [34]. A blockchain and NFT-based solution has addressed the challenges of managing, sharing, and monetizing genomic data with privacy preserving techniques [35].

B. GENERATIVE ADVERSARIAL NETWORKS

A Generative Adversarial Network (GAN) is a machine learning model, widely used in solving several computer vision tasks [36]. The primary idea behind GAN is to train two neural networks, namely a generator and a discriminator. Here's a high-level overview of how GANs work in the context of image processing.

Generator (G): The generator function takes random noise $\mathbf{z}$ as input and produces synthetic data, representing an image $\mathbf{x}_{fake}$. The GAN generator function is defined as

$$\mathbf{x}_{fake} = G(\mathbf{z}; \theta_G) \quad (1)$$

where:

- $\mathbf{z}$ is the random noise taken as input
- θ_G represents the parameters of the generator network

The generator function, denoted as $G(\mathbf{z}; \theta_G)$, given in eq. 1. Initially, the generator produces low-quality images, as it starts with random weights.

Discriminator (D): The discriminator network is trained to distinguish between real and fake (generated) data. Discriminator (D) takes an input x and outputs a probability $D(x)$ indicating the likelihood that an input x is a real image. The discriminator function is denoted as $D(x; \theta_D)$, given as

$$D(\mathbf{x}_{real}) = D(x; \theta_D) \quad (2)$$

where:

- x can either be a real image or a generated image
- θ_D represents the parameters of discriminator network

Training Process: The generator and discriminator are trained iteratively in an adversarial (competitive) process. The generator aims to improve its ability to generate realistic images to fool the discriminator. This is equivalent to maximizing the objective function (eq. 3). Whereas, the discriminator tries to perform better in distinguishing between the real and fake images (eq. 4). The objective function is given as

$$\max V(G, D)_{\theta_G} = \log(1 - D(G(\mathbf{z}))) \quad (3)$$

where:

- $V(G, D)$ represents generator objective function
- $\max V(G, D)_{\theta_G}$ represents maximization over the parameters (θ) of the Generator (G).

$$\max V(D, G)_{\theta_D} = \log(D(x_{real})) + \log(1 - D(G(\mathbf{z}))) \quad (4)$$

where:

- $V(D, G)$ represents discriminator objective function
- $\max V(D, G)_{\theta_D}$ represents maximization over the parameters (θ) of the Generator (G).

The generator and discriminator are optimized using a loss function. The generator's objective is to maximize the probability of the generated images are classified as real by the discriminator. Whereas, the discriminator's objective is to correctly classify the real and fake images. Ideally, as training progresses, the generator becomes skilled at generating images that are indistinguishable from real ones, and the discriminator becomes less effective at telling them apart. The objective function for adversarial training is defined (eq. 5) as

$$\min_{\theta_G} \max_{\theta_D} V(G, D) = \mathbb{E}_{x_{real}} [\log(D(x_{real}))] + E_{noise} \quad (5)$$

$$E_{noise} = \mathbb{E}_z [\log(1 - D(G(\mathbf{z}))) \quad (6)$$

where:

- $V(G, D)$ represents overall objective function
- $E_{x_{real}}$ denotes the expectation over real distribution
- E_z is the expectation over noise vector distribution.

GANs have been successfully used for various image processing tasks, such as image synthesis, style transfer, image-to-image translation, super-resolution, and many more. GANs can also be used for data augmentation, creating diverse datasets for training other machine learning models. In this context, NFTGAN proposed a GAN-based model for NFT-style digital art generation [37].

C. IMAGE PROCESSING BASED ACCOUNTABILITY

Several multiplexing techniques such as spatial multiplexing and frequency-domain multiplexing have been explored to investigate their efficacy in image superimposition and encryption [38], [39]. Image encryption usually involves transforming pixel values through mathematical operations or substituting pixels with encrypted counterparts. Multiplexing introduces a concept of combining RGB values from different images, thereby increasing intricacy of an encryption process. This approach holds the potential to thwart unauthorized access and enhance resilience of encrypted images against various attacks [40]. Spatial multiplexing involves combining RGB values at pixel-level, while frequency-domain multiplexing leverages transformations, like Fourier analysis to merge images in frequency-domain [41].

The ubiquitous use of digital imagery necessitates advanced techniques for securing visual data against unauthorized access and tampering. GANs known for their ability to generate realistic images, offer a unique avenue for exploring novel methods in image superimposition and encryption. Alpha-blending involves combining two images by assigning each pixel a transparency value (alpha) that determines its contribution to the final output. Advanced Encryption Standard (AES) [42] is a symmetric encryption algorithm. Its adoption in image processing contributes to creating a secure environment for transmitting, storing, and sharing visual information, making it a cornerstone in the realm of digital content protection.

Gaussian noise [22] is a statistical noise characterized by a probability density function (PDF) that follows a Gaussian (normal) distribution. In the context of image

processing, it manifests as random variations in pixel values, introducing a subtle, continuous randomness to the image. Fast non-linear image blending [43] involves combining multiple images in a seamless and visually appealing manner while incorporating non-linear operations. This technique is useful in applications such as panoramic image stitching, exposure blending, special effects in computer graphics and related domains.

D. IMAGE ANALYSIS: MATCHING SIMILARITY AND QUALITY METRICS

Fast Library for Approximate Nearest Neighbors (FLANN) matching in image analysis has sparked significant interest due to its ability to boost matching precision and speed. Recent studies, shed light on the power of FLANN in pairing feature points efficiently [44], [45]. Scale-Invariant Feature Transform (SIFT) is an image descriptor used for identifying and describing distinct features of an image. Combination of SIFT with FLANN and RANSAC (RANdom SAMple Consensus) showcased superior real-time image recognition and matching compared to traditional methods [46].

Previous research emphasized the strengths of SIFT, SURF, and PCA-SIFT in various contexts, underlining the adaptability and performance nuances of each algorithm [23], [24]. FLANN's versatility in tandem with descriptors like SIFT and SURF, utilizing diverse distance metrics and Lowe's ratio test for match filtering, continues to drive advancements in image matching algorithms. Ongoing investigations aim to optimize FLANN's synergy with different feature descriptors, promising further enhancements in image matching [48].

Pixel-level cosine similarity measures the similarity based on raw pixel intensities, treating images as high-dimensional vectors in RGB color space. However, it may be sensitive to variations in illumination and minor distortions, limiting its effectiveness in scenarios where semantic content is crucial [49]. In contrast, feature-based cosine similarity utilizes convolutional neural networks (CNNs) to extract high-level features from images [50]. By considering activations of intermediate layers as feature vectors, this approach captures rich semantic information, offering superior robustness in illumination and viewpoint variations.

It is critical to measure the perceptual and structural differences between images in the fields of computer vision and image processing [51]. Structural Similarity Index (SSIM) and Peak Signal-to-Noise Ratio (PSNR) are two widely used measures for this purpose. The SSIM, proposed by Wang et al. in 2004 [52], assesses the similarity between two images based on three components: luminance, contrast, and structure. It is defined as

$$\text{SSIM}(x, y) = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (7)$$

where

- x and y are the two images being compared.
- μ_x and μ_y are the averages of x and y .

- σ_x^2 and σ_y^2 are the variances of x and y .
- σ_{xy} is the covariance of x and y .
- $C_1 = (K_1L)^2$ and $C_2 = (K_2L)^2$ are two variables to stabilize the division with weak denominator, where L is the dynamic range of the pixel values (typically this is 255 for 8-bit grayscale images), and K_1 and K_2 are constants (default values are $K_1 = 0.01$ and $K_2 = 0.03$).

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\text{MSE}} \right) \quad (8)$$

where:

- L is the maximum possible pixel value of the image (for an 8-bit image, L is 255).
- MSE is the mean squared error between the original image and the distorted image.

The mean square error (MSE) is calculated as:

$$\text{MSE} = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2 \quad (9)$$

where I is an original image, a pixel is $I(i, j)$, K is a distorted image, and m and n are the row and column dimensions of the images, respectively.

The SSIM provides a comprehensive measure of perceptual similarity that is highly aligned with human visual perception by taking into account local patterns in these components. The SSIM ranges within ± 1 , where a value closer to 1 indicates higher similarity. This method excels at capturing informative details from both the entire image (global image feature) and specific areas (local image feature) within it. This makes it resistant to common image imperfections like noise and compression artifacts.

In contrast, robustness metrics like Peak Signal-to-Noise Ratio (PSNR) and Bit Error Rate (BER) also play pivotal roles in the image-based evaluation process. The PSNR measures the ratio between the maximum possible power of a signal and the power of corrupting noise. It is represented in Decibels (dB). It offers a simple numerical indicator of the reconstruction quality by comparing pixel values of original and distorted images [53]. A higher PSNR value implies a lesser distortion is present, hence PSNR having high values represent excellent quality. The values are greater than zero. PSNR's simplicity offers a clear metric, but it falls short when considering human perception. However, it remains valuable for applications needing precise numerical comparisons.

Numerous studies have explored the efficacy of SSIM and PSNR in various contexts. Smith and Brady [54] demonstrated the superiority of SSIM over PSNR in assessing image quality in the presence of Gaussian blur and additive white Gaussian noise. Wang et al. proposed multi-scale SSIM to enhance the original SSIM formulation for further improving the performance [55]. Other data sharing models like RCDS discussed security issues in blockchain data transfer.

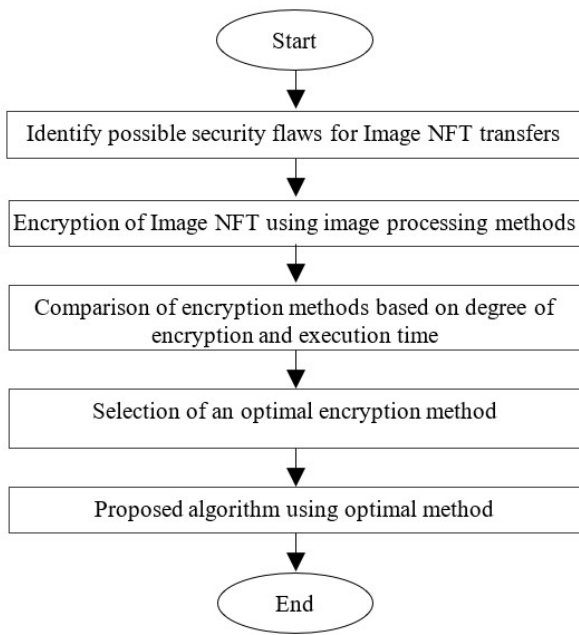


FIGURE 1. Flowchart represents methodology of this paper.

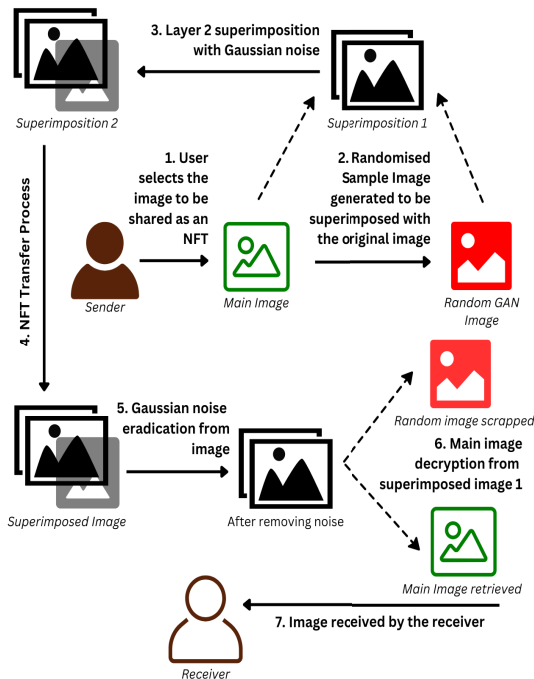


FIGURE 2. High level user architecture representing the proposed methodology.

III. PROPOSED METHODOLOGY

The implemented NFT transactions via Ethereum blockchain are studied from previous literature, and security risks in image NFT transfers are identified as notable research gaps. An algorithm is devised to add an encryption layer to image NFTs during transfer. This technique is independent of NFT markets or wallets used for transaction. An encrypted image metadata is sent via Ethereum blockchain (on-chain) and

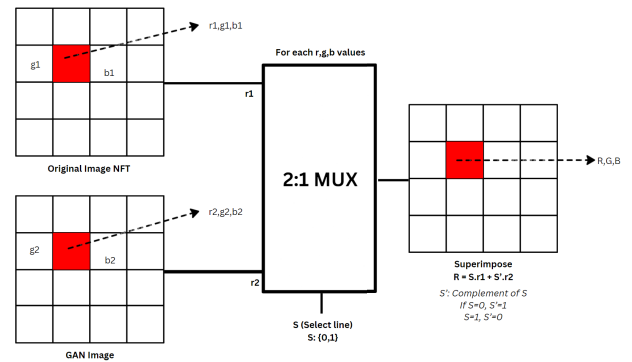


FIGURE 3. Multiplexing 'red' values of same pixel position in Original Image NFT and GAN Image to produce Superimposed Image.

some of the decryption data off-chain. At any point, neither the user, nor the system has enough data to obtain the entire image NFT metadata, thus reducing the security vulnerability. A schematic representation of the proposed methodology is shown in Fig. 1.

Multiple pixel-based algorithms are compared based on their level of encryption and execution times, and an optimal algorithm is selected. The FLANN matching technique is used for image matching of the original and encrypted images to compare the levels of encryption. Pixel-based encryption algorithms and FLANN matcher have been discussed. Cosine similarity, SSIM and PSNR comparisons are discussed to verify and corroborate the FLANN matching results.

The optimal image processing technique has been used in the proposed algorithm. A high-level overview has been represented in Fig. 2, describing the proposed image-based NFT transfer technique. The proposed algorithm uses application of GANs along with Gaussian noise in alpha composition with an original image NFT. Alpha composition is utilized as a mechanism to embed encryption features within the blended images while GANs may enhance the security and robustness in the encryption process.

The encryption algorithms based on image processing techniques have been represented ahead. Algorithm 1 corresponds to an encryption method using custom bit-wise operations applied to RGB values. Algorithm 2 refers to alpha blending, whereas Algorithm 3 describes addition of Gaussian noise. Algorithms 4 and 5 describe the encryption and decryption techniques implemented for Advanced Encryption Standard (AES) method. Our proposed encryption and decryption methods are ideated in Fig. 2 - Fig. 6 subsequently.

A. ENCRYPTION METHOD

1) RGB MULTIPLEXING

The composition of an image $I_{height \times width \times 3}$ is determined by its pixel values. RGB multiplexing involves manipulating color components and thus is a determining factor in distinguishing between images. A previous study indicated use of fractional Fourier transform technique to encrypt four RGB images into one using elementary multiplexing

Algorithm 1 RGB Encryption Algorithm

```

function RGB Encryption ( $I_{height \times width \times 3}$ )
  Initialize: 3D array  $A_{height \times width \times 3} \leftarrow 0$ ,
  alpha-channel  $\alpha \leftarrow 0$ , and constant  $k = 50$ 
  1. for  $i = 1$ : height do
  2.   for  $j = 1$ : width do
  3.     $(r, g, b, a) \leftarrow \text{getPixelValues}(I_{i \times j \times 3})$ 
  4.     $P \leftarrow k \times (i \oplus j)$ 
  5.     $r \leftarrow (((P \oplus r) - g) \oplus b)$ 
  6.     $g \leftarrow (((P \oplus g) - b) \oplus r)$ 
  7.     $b \leftarrow (((P \oplus b) - r) \oplus g)$ 
  8.    if  $r > 255$  then
  9.      $A_{i \times j \times 0} \leftarrow \lfloor r/255 \rfloor$ 
  10.     $r' \leftarrow r \% 255$ 
  11.   end if
  12.   if  $g > 255$  then
  13.     $A_{i \times j \times 1} \leftarrow \lfloor g/255 \rfloor$ 
  14.     $g' \leftarrow g \% 255$ 
  15.   end if
  16.   if  $b > 255$  then
  17.     $A_{i \times j \times 2} \leftarrow \lfloor b/255 \rfloor$ 
  18.     $b' \leftarrow b \% 255$ 
  19.   end if
  20.    $I'_{i \times j \times 3} \leftarrow \text{setPixelValues}(A_{i \times j \times (r', g', b')})$ 
  21.  end for
  22. end for
  23. return  $I'_{i \times j \times 3}$ 
end function

```

Algorithm 2 Alpha Blending

```

function Alpha_Blending( $I_1, I_2, \alpha$ )
  1.  $I_1 \leftarrow \text{ReadImage}(I_1)$ 
  2.  $I_2 \leftarrow \text{ReadImage}(I_2)$ 
  3.  $I_2 \leftarrow \text{ResizeImage}(I_2, \text{width} = I_1.\text{width}, \text{height} = I_1.\text{height})$ 
  4.  $I_{\text{superimposed}} \leftarrow \text{CreateEmptyImage}(\text{width} = I_1.\text{width}, \text{height} = I_1.\text{height})$ 
  5. for each pixel  $(x, y)$  in  $I_1$  and  $I_2$  do
  6.    $RGB_1 \leftarrow \text{GetRGBvalue}(I_1, x, y)$ 
  7.    $RGB_2 \leftarrow \text{GetRGBvalue}(I_2, x, y)$ 
  8.    $RGB_{\text{blend}} \leftarrow \text{BlendRGB}(RGB_1, RGB_2, \alpha)$ 
  9.    $\text{SetRGBvalue}(I_{\text{superimposed}}, x, y, RGB_{\text{blend}})$ 
  10. end for
  11. return  $I_{\text{superimposed}}$ 
end function

```

operations [56]. Multiplexing is long known for encryption of values to hide them from alien players. RGB multiplexing represents a mixture of RGB values of exactly equivalent-dimensional images, referring to equal places of RGB. One can involve a select line metric of any choice. Higher levels of encryption may be attained by involving neighboring pixel values. As the neighbor pixels may be used furtively, it makes

Algorithm 3 Addition of Gaussian Noise

```

1. function Add_Gaussian_Noise( $I, \text{stdDeviation}$ )
2.    $\text{noise} \leftarrow \text{Random}(0, \text{stdDeviation}, \text{sizeOfImage})$ 
3.    $I_{\text{noise}} \leftarrow \text{AlphaBlending}(I, \text{noise})$ 
4.    $I_{\text{noise}} \leftarrow \text{ClipValues}(I_{\text{noise}}, 0, 255)$ 
5.   return  $I_{\text{noise}}$ 
6. end function

```

Algorithm 4 AES_IMAGE_ENCRYPT

```

function AES_Img_Encrypt(image, key)
  1.  $\text{rows}, \text{cols} \leftarrow \text{Dimensions of the image}$ 
  2.  $\text{flat\_image} \leftarrow \text{Flatten}(\text{image})$ 
  3.  $\text{blocks} \leftarrow \text{SplitIntoBlocks}(\text{flat\_image}, 16)$ 
  4.  $\text{expanded\_key} \leftarrow \text{KeyExpansion}(\text{key})$ 
  5.  $\text{encrypted\_blocks} \leftarrow []$ 
  6. for each block in blocks do
  7.    $\text{state} \leftarrow \text{AddRoundKey}(\text{block}, \text{expanded\_key}[0 : 16])$ 
  8.   for round = 1 to  $Nr - 1$  do
  9.     $\text{state} \leftarrow \text{SubBytes}(\text{state})$ 
  10.    $\text{state} \leftarrow \text{ShiftRows}(\text{state})$ 
  11.    $\text{state} \leftarrow \text{MixColumns}(\text{state})$ 
  12.    $\text{state} \leftarrow \text{AddRoundKey}(\text{state}, \text{expanded\_key}[\text{round} \times 16 : (\text{round} + 1) \times 16])$ 
  13.  end for
  14.   $\text{state} \leftarrow \text{SubBytes}(\text{state})$ 
  15.   $\text{state} \leftarrow \text{ShiftRows}(\text{state})$ 
  16.   $\text{state} \leftarrow \text{AddRoundKey}(\text{state}, \text{expanded\_key}[\text{Nr} \times 16 : (\text{Nr} + 1) \times 16])$ 
  17.   $\text{encrypted\_blocks} \leftarrow \text{Append}(\text{encrypted\_blocks}, \text{state})$ 
  18. end for
  19.  $\text{encrypted\_image} \leftarrow \text{Reshape}(\text{encrypted\_blocks}, \text{rows}, \text{cols})$ 
  20. return  $\text{encrypted\_image}$ 
end function

```

unauthorized decryption to be difficult. A sample case of multiplexing 2-pixel positions for R (red) value is shown in Fig. 3. Select lines can be used as a mixture of any neighboring pixel values and kept hidden.

A custom encryption algorithm based on using bit-wise XOR operation on RGB components is shown in Algorithm 1. This algorithm takes a single 2D image input ($I_{height \times width \times 3}$) and a constant value K , which may be used to add attribute dependency later. Each pixel in the 2D image has $\{R, G, B\}$ components and is accessed using the width and height values. For each pixel, the RGB components are encrypted based on the custom functions. In Algorithm 1, if either of these pixel component values exceed 255, then the

Algorithm 5 AES_IMAGE_DECRYPT

```

function AES_Img_Decrypt(encrypt'd_img, key)
1. rows, cols ← Dimensions of encrypted_image
2. flat_encrypted_img ← Flatten(encrypted_img)
3. blocks ← SplitIntoBlocks(flat_encrypt'd_img, 16)
4. expanded_key ← KeyExpansion(key)
5. decrypted_blocks ← [ ]
6. for each block in blocks do
7. state ←
   AddRoundKey(block, expanded_key[ Nr × 16 :
   (Nr + 1) × 16])
8. state ← InvShiftRows(state)
9. state ← InvSubBytes(state)
10. for round = Nr − 1 to 1 do
11. state ←
   AddRoundKey(state, expanded_key[round ×
   16 : (round + 1) × 16])
12. state ← InvMixColumns(state)
13. state ← InvShiftRows(state)
14. state ← InvSubBytes(state)
15. end for
16. state ←
   AddRoundKey(state, expanded_key[0 : 16])
17. decrypted_blocks ←
   Append(decrypted_blocks, state)
18. end for
19. decrypted_image ←
   Reshape(decrypted_blocks, rows, cols)
20. return decrypted_image
end function

```

value is divided by 255 and the quotient is stored separately in an array for decryption, while the remainder becomes the pixel component value.

2) ADVANCED ENCRYPTION STANDARD (AES)

It is a symmetric encryption algorithm widely used to secure sensitive data [57], and is used to encrypt and decrypt images to protect their confidentiality. The AES uses a secret key for encryption and decryption. Thus, the key must be kept secure and shared only between entities requiring an access permission to the encrypted images.

Algorithms 4 and 5 describe AES encryption and decryption on a sample input image. During encryption, these processes alter image data significantly, which is later restored in decryption. A secure key is selected and is converted into an expanded key. The image data is divided into blocks and combined with the expanded key using the 'AddRoundKey' step. Further, these image blocks are substituted, and the rows and columns within blocks have been rearranged. This results in an encrypted image data block output, which can be decrypted back into the original image using the expanded key and inverse operations. This transformation ensures that the image data remains

secure during transmission or storage, as the encrypted data provides no clue about the original image without the correct decryption key.

3) ALPHA BLENDING

it is a technique to combine two images or layers together in a way that allows transparency. The term *alpha* refers to alpha channel, an additional channel in an image that represents the transparency or opacity of each pixel. The alpha channel values range from 0 (i.e., completely transparent) to 1 (i.e., completely opaque). In image superimposition, alpha blending enables an overlaying of one image onto another while adjusting transparency levels. This creates a smooth and visually appealing transition between the two source images.

In Algorithm 2, alpha blending of two images (I_1 and I_2) has been discussed. The images are resized to the same number of pixels in both images. For each pixel, the RGB values are extracted separately from each input image. The RGB values are then blended using the alpha factor to create a new RGB value for generating the superimposed image $I_{superimposed}$. The blending formula for each channel $ch \in \{R, G, B\}$ is shown in Eq. 10. However, alpha blending is still primitive in terms of complex image superimposition.

$$I_{blended} = \alpha \times I_1^{ch} + (1 - \alpha) \times I_2^{ch} \quad (10)$$

4) GENERATIVE ADVERSARIAL NETWORK (GAN)

it is successfully used for various image processing tasks, such as image synthesis, style transfer, image-to-image translation, and super-resolution. GANs can also be used for data augmentation, creating diverse datasets for training other machine learning models. In context of image encryption, GANs can be used to randomly generate an input image using RGB multiplexing or alpha blending.

5) GAUSSIAN NOISE

it is a type of random noise that can be added to images for various purposes, like simulating real-world conditions or testing the robustness of image processing algorithms. The process involves adding random values drawn from a Gaussian distribution to pixel values of an image. Insertion of Gaussian noise is discussed in Algorithm 3. The image blended with GAN is taken as input along with standard deviation for noise. The input image is blended with random noise based on the standard deviation and pixel component values are clipped to valid range. Thus, a noisy image I_{noise} is obtained as output which may be decrypted by reversing the blending process. Sample outputs of the proposed method are illustrated in Fig. 11 - 12.

B. PROPOSED ENHANCED ENCRYPTION-DECRYPTION ALGORITHM

Our work employs various image processing techniques, including image loading, encryption, RGB multiplexing, and alpha composition (with AES). Additionally, GANs

Algorithm 6 Enhanced Encryption-Decryption Algorithm - Functions

```

function SHA256Hash (image)
1.  image_data  $\leftarrow$  ReadImageData(image)
2.  hash_value  $\leftarrow$  SHA256(image_data)
3.  return hash_value
end function
function AlphaBlend(img1, img2, alpha)
4.  OUTPUT_IMAGE  $\leftarrow$  CreateEmptyImage(SameSizeAs(img1))
5.  for each pixel in img1 do
6.    OUTPUT_IMAGE[pixel]  $\leftarrow$  (img1[pixel]  $\times$  alpha) + (img2[pixel]  $\times$  (1 - alpha))
7.  end for
8.  return OUTPUT_IMAGE
end function
function EncryptBMD(Alpha1, Alpha2, NFTImage, GANImage, Noise)
9.  metadata  $\leftarrow$  []
10. BlendImage  $\leftarrow$  AlphaBlend(NFTImage, GANImage, Alpha1)
11. EncryptedImage  $\leftarrow$  AlphaBlend(BlendImage, Noise, Alpha2)
12. metadata["SentImage"]  $\leftarrow$  SHA256Hash(EncryptedImage)
13. metadata["Alpha1"]  $\leftarrow$  Alpha1 ; metadata["Alpha2"]  $\leftarrow$  Alpha2
14. return metadata
end function
function SendMetadataOnchain(metadata, SBA, RBA)
15. TRANSACTION_ID  $\leftarrow$  SendToBlockchain(metadata, SBA, RBA)
16. Return TRANSACTION_ID
end function
function SENDDATAOFFCHAIN(GANImage, Noise, SA, RA)
17. OFF_CHAIN_DATA  $\leftarrow$  SendImages(GANImage, Noise, SA, RA)
18. return OFF_CHAIN_DATA
end function
function ReceiveMetadataOnchainTRANSACTION_ID, RBA
19. return GetFromBlockchain(TRANSACTION_ID, RBA)
end function
function RECEIVEDATAOFFCHAIN(offchaindata, RA)
20. GANImage, Noise  $\leftarrow$  GetImages(offchaindata, RA)
end function
function ReverseAlphaBlend(img1, img2, alpha)
21. OUTPUT_IMAGE  $\leftarrow$  CreateEmptyImage(SameSizeAs(img1))
22. for each pixel in img1 do
23.   OUTPUT_IMAGE[pixel]  $\leftarrow$  (img1[pixel] - (1 - alpha)  $\times$  img2[pixel]) / alpha
23. end for
24. return OUTPUT_IMAGE
end function
25. function DecryptBMD(metadata, GANImage, Noise)
26. metadata["ReceivedImage"]  $\leftarrow$  SHA256Hash(EncryptedImage)
27. Alpha1  $\leftarrow$  metadata["Alpha1"]; Alpha2  $\leftarrow$  metadata["Alpha2"]
28. if metadata["ReceivedImage"] == metadata["SentImage"] then
   /* Image integrity verified */
30. else /* Image integrity compromised */
31. end if
32. Blending2  $\leftarrow$  ReverseAlphaBlend(EncryptedImage, GANImage, Alpha1)
33. DecryptedImage  $\leftarrow$  ReverseAlphaBlend(Blending2, Noise, Alpha2)
34. return DecryptedImage
end function

```

Algorithm 7 Enhanced Encryption-Decryption Algorithm: Main()

```

function FullProcess
1. INPUTS: Alpha1, Alpha2, Noise, GANImage, NFTImage, SBA, SA, RBA, RA
   // ENCRYPTION //
2. metadataSent  $\leftarrow$  EncryptBMD(Alpha1, Alpha2, NFTImage, GANImage, Noise)
   //SENDER SENDS DATA //
3. transactionId  $\leftarrow$  SendMetadataOnchain(metadataSent, SBA, RBA)
4. offchainData  $\leftarrow$  SENDDATAOFFCHAIN(GANImage, Noise, SA, RA)
   // RECEIVER GETS DATA //
5. metadataReceived  $\leftarrow$  ReceiveMetadataOnchain(transactionId, RBA)
6. RECEIVEDATAOFFCHAIN(offchainData, RA)
   // DECRYPTION //
7. OUTPUT_IMAGE  $\leftarrow$  DecryptBMD(metadataReceived, GANImage, Noise)
8. print: TRANSACTION COMPLETED!
end function

```

and Gaussian noise are used to compare encrypted image outputs based on execution time and encryption levels. This comparison is depicted in Fig. 7 and Fig. 8, showcasing the encryption methods and their comparisons. In the proposed Algorithm-6 and 7 for the transfer and decryption of image NFTs via blockchain, a comprehensive process is outlined.

- **SHA256HASH** function initiates by reading an input image (I) and computing its hash-value.

- **ALPHABLEND** function blends two images using a specified alpha value, calculating a weighted sum of pixel values.

- **ENCRYPTBMD** function encrypts the blockchain meta-data by blending the NFT image (INFT) and a GAN image (IGAN), followed by blending with noise, and computes the hash for integrity verification.

- **SBA** denotes Sender Blockchain Address, **RBA** denotes Receiver Blockchain Address, **SA** denotes Sender Address (OffChain), **RA** denotes Receiver Address (RA). This metadata, including alpha values and hashes, is sent to the blockchain, via the **SENDMETADATAONCHAIN** function, which securely transmits it using sender and receiver blockchain addresses.

- **RECEIVEMETADATAONCHAIN** function retrieves and decrypts this metadata, ensuring secure transmission between sender and receiver addresses. Subsequently, **DECRYPTBMD** function decrypts the data by performing inverse blending operations using the provided metadata.

The proposed Algorithms 6-7 effectively utilize cryptographic hashing, image blending, and blockchain technology to ensure secure transfer, encryption, and verification of image NFTs.

1) IMAGE ENCRYPTION

In this primary phase, the proposed algorithm performs alpha composition of original image NFT with a randomly generated image leveraging GAN. Subsequently, Gaussian noise is introduced to complement GAN image and boost robustness of the encrypted image metadata [55]. As complexity of

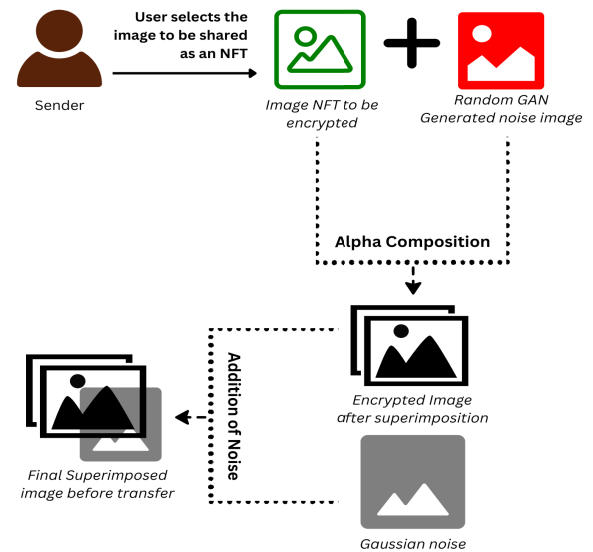


FIGURE 4. Encryption technique of the proposed algorithm.

model grows, the execution time increases, providing hackers more time to attack. Alpha blending allows higher levels of encryption of images in lower execution times, and thus is a preferred solution. Randomly generated GAN images infused with Gaussian errors offer fast and secure encryption. Fig. 4 depicts image encryption method adopted in the proposed algorithm. Overall encryption strategy via on-chain and off-chain transfers is shown in Fig. 5.

2) NFT TRANSFER PHASE

Once the image is encrypted, its Secure Hash Algorithm 256-bit (SHA256) version metadata is shared over blockchain. For 0-value NFTs, the NFTs with no price in cryptocurrency, the transfer may be completed without spending gas fees over marketplaces, like OpenSea, Rarible, Larva, etc. Numerous risks are associated with different NFT marketplaces. This model does not cover all associated risks

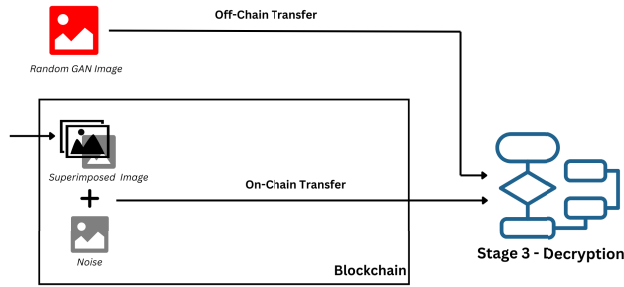


FIGURE 5. Transfer step of the proposed algorithm.

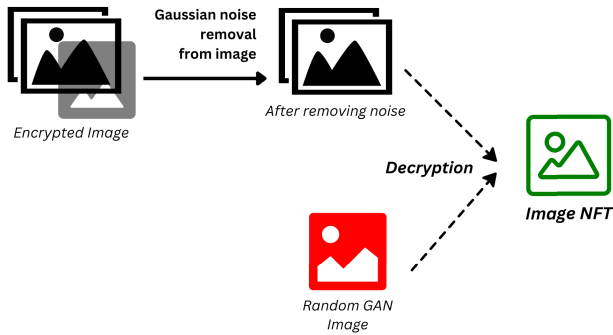


FIGURE 6. 2-step decryption for proposed algorithm.

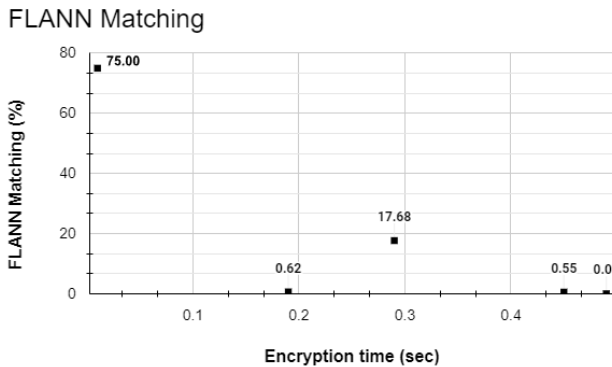


FIGURE 7. Graph representation of FLANN Matching, Percentage Values lie between 0-100.

across these platforms and the possible attack vectors have been discussed next. However, theft cases can get mitigated as encrypted image metadata is sent over the blockchain, thus, double-proofing the image NFT. While image encryption adds an extra layer of security, the overall security of NFT transfer process relies on secure wallet practices, robust blockchain networks, and trustworthy NFT marketplaces. Regular audits, monitoring, and adherence to best security practices are essential to mitigate potential risks associated with the NFT marketplace stage. It is crucial to stay informed about evolving security threats and adapt security measures accordingly.

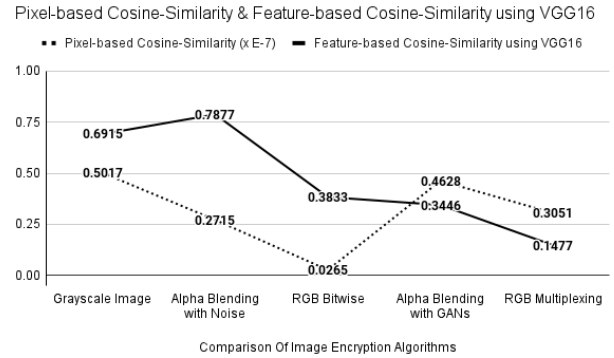


FIGURE 8. Graph representation of Pixel based Cosine & Feature based Cosine Similarity using VGG16. All values lie between 0.0-1.0.

3) IMAGE DECRYPTION PHASE

This phase focuses on the decryption of the received metadata post the completion of NFT transfer. A two-step decryption process is performed to remove Gaussian noise and further retrieve the superimposed image from alpha blending. It is illustrated in Fig. 6, where the GAN image is sent off-chain as shown in Fig. 5. The first part of decryption is noise removal done using metadata sent on-chain of encrypted image and noise. Alpha blending process is reversed using the GAN image after noise removal.

IV. RESULT ANALYSIS AND DISCUSSIONS

From previous literature, current NFT transfer systems were assessed and some potential risks and security vulnerabilities were identified. In particular, image NFTs are of high economic values and more susceptible to thefts. Analysis of image NFT transactions and case studies have been discussed, revealed presence of vulnerabilities in smart contracts. Attackers may utilize the fact that an original image NFT metadata is available during transfer via the smart contracts. An additional encryption protection scheme in the algorithm is proposed to mitigate the security risk of losing original image NFT to attackers. The algorithm employs an optimized encryption technique such that the encrypted image metadata will be transferred instead. For the decryption, some metadata may be sent off-chain while the rest via on-chain i.e., blockchain. This would ensure that no unauthorized entity has access to entire encrypted image NFT metadata. If an attacker wants to gain access, it is needed to collect and decode both off-chain and on-chain metadata simultaneously in a lesser time span than the current time.

A. COMPARISON OF IMAGE PROCESSING TECHNIQUES

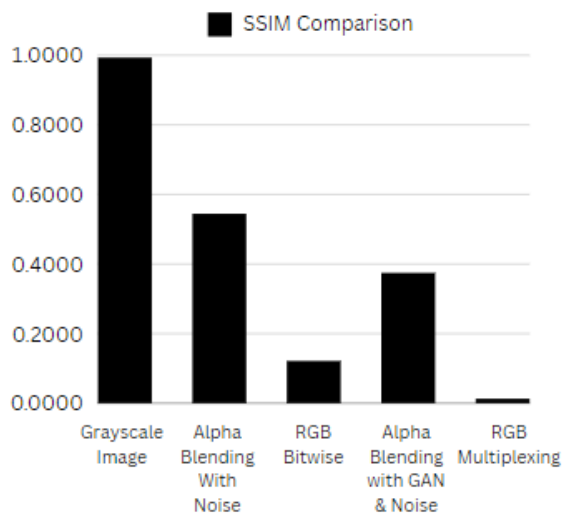
Based on the FLANN matching output, the encryption matches less than nearly 1% for RGB operations or alpha blending technique combined with Gaussian noise and GANs (see Table 1). Thus, these are better choices for image encryption. However, due to existence of AI-based algorithms designed for decryption, simple RGB bit-wise or

TABLE 1. A comparison of image processing techniques, their levels of encryption and execution times using FLANN matching.

Image processing techniques	FLANN matching of encrypted and original images	Encryption time
Grayscale image	75 %	0.01 seconds
Alpha blending with noise	17.68 %	0.29 seconds
RGB bitwise	0.62 %	0.19 seconds
Enhanced Encryption-Decryption Algorithm	0.55 %	0.45 seconds
RGB multiplexing	0.01 %	0.49 seconds

TABLE 2. A comparison of image processing techniques based on pixel-based cosine-similarity and Feature-based cosine-similarity using VGG-16 model.

Image processing techniques	Pixel-based cosine-similarity	Feature-based cosine-similarity using VGG-16
Grayscale image	5.0172e-08	0.6915
Alpha blending with noise	2.7152e-08	0.7877
RGB bitwise	2.6476e-09	0.3833
Enhanced Encryption-Decryption Algorithm	4.6276e-08	0.3446
RGB multiplexing	3.0514e-08	0.1477

**FIGURE 9.** SSIM comparison of Image encryption algorithms.

multiplexing operations are not preferred choices for image NFT encryption.

GANs provide randomly generated images, which are alpha blended with image NFTs and can be transferred in a fast and secure way. Introduction of Gaussian noise further improves security, making it harder for attackers to get access to original image NFT.

The comparison of generated image processing output images has been performed using SSIM and PSNR techniques, discussed in Table 2. The SSIM index values range from -1 to 1 with value being closer to 1 for higher similarity. Higher PSNR values indicate higher similarity between the images, while lower PSNR values indicate lower similarity. A greater PSNR value indicates that the images are more similar in terms of pixel-wise fidelity. The results in Table 2 for various image processing techniques resonate with the observations obtained from FLANN matching.

The pixel-based cosine similarity and the feature based cosine similarity using VGG-16 deep learning model are

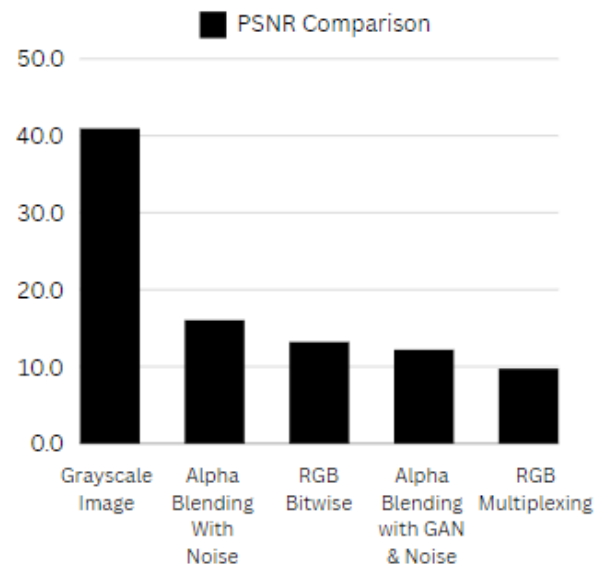
**FIGURE 10.** PSNR (db) comparison of image encryption algorithms.**TABLE 3.** A comparison of image processing techniques based on the SSIM and PSNR techniques.

Image processing techniques	SSIM	PSNR (db)
Grayscale image	0.9932	40.9397
Alpha blending with noise	0.5444	16.0345
RGB bitwise	0.1215	13.1923
Alpha blending with Gaussian noise and GANs	0.3745	12.1886
RGB multiplexing	0.0128	9.7266

compared and discussed in Table 3. Higher cosine similarity values are indicative of increased likeness between the images, yet it is essential to interpret the results in the context of specific application and the characteristics of images being compared. While pixel-based cosine similarity centers on the direct comparison of raw pixel values, feature-based cosine similarity utilizes information gleaned from the VGG-16 to encapsulate semantic content. Consequently, feature-based similarity may offer a more nuanced assessment concerning image content and semantics.

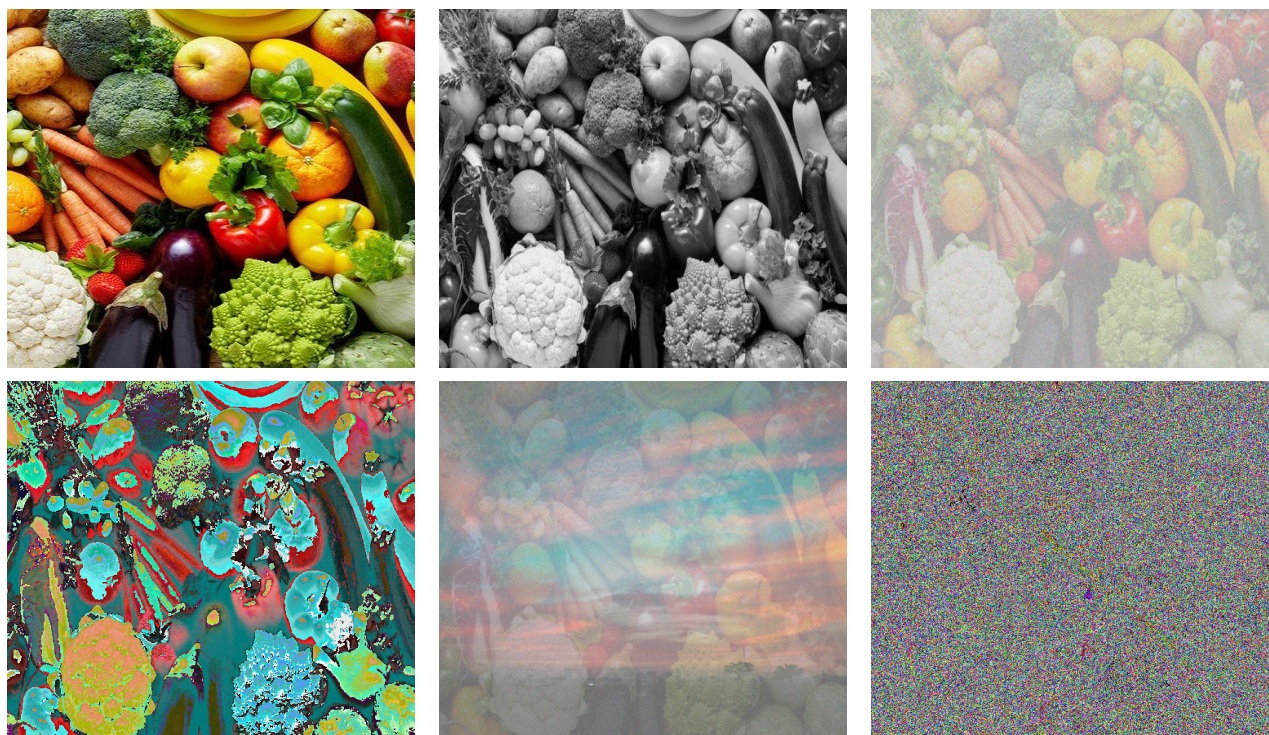


FIGURE 11. The sample input image and its corresponding output images obtained as Grayscale image, alpha blending with noise, RGB bitwise encryption, alpha blending using GANs and noise, RGB multiplexing technique.

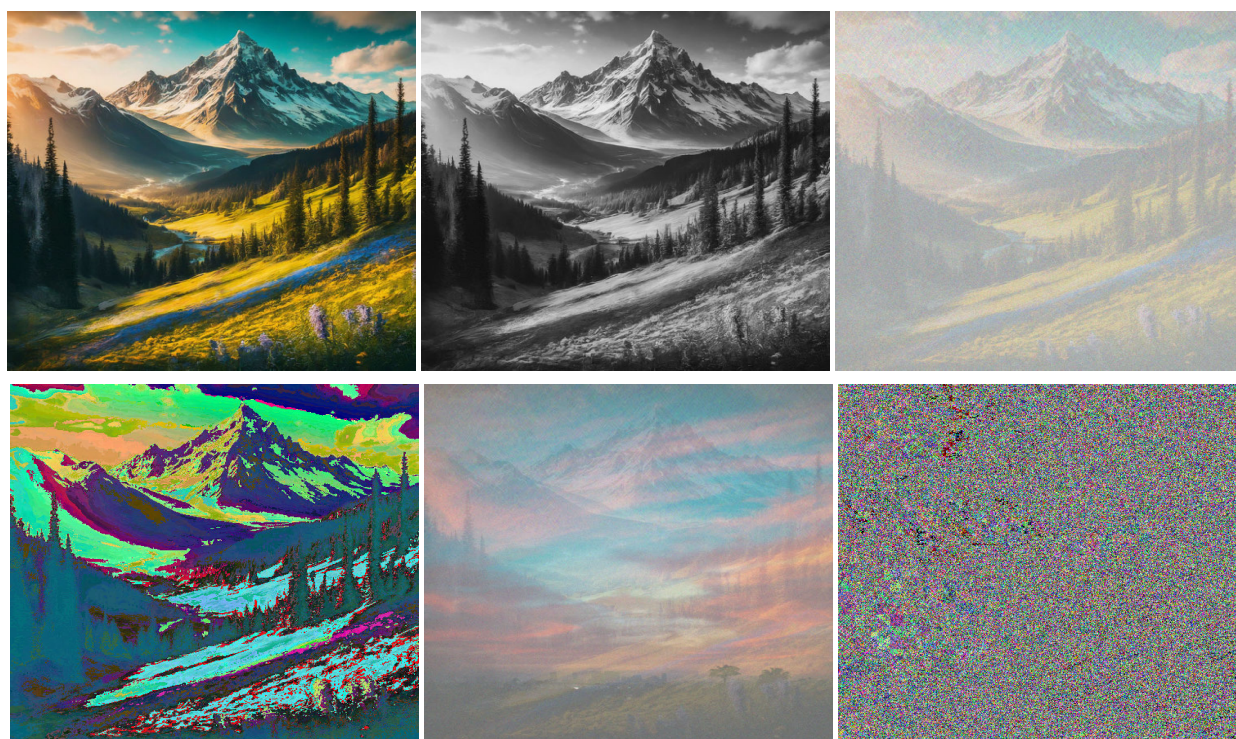


FIGURE 12. The sample input image 2 and its corresponding output images obtained as Grayscale image, alpha blending with noise, RGB bitwise encryption, alpha blending using GANs and noise, RGB multiplexing technique.

B. DESIGN CHOICES AND ADVANTAGES

1) USING A BLOCKCHAIN IN THE SYSTEM ARCHITECTURE

Traditional media transfer takes place over a cloud or via physical sharing. Blockchain makes the process bulletproof

for a user by encrypting the image asset into a NFT. It makes a visual picture into hash (metadata) which is sent over a blockchain, thereby double proofing our engine mechanism.

2) CHOICE OF ENCRYPTION ALGORITHM

In the past decade, various avenues have opened up for image processing, especially in the domain of security using superimposition. In the near future, AI will be used to create random adversarial images which will always be different from previously used. GANs create slightly different images, and thus once performed with Gaussian noise make a transfer robust. Hence, it will have absolute unique RGB, and difficult to hack/attack.

3) LAYERED SECURITY USING MULTIPLE ALGORITHMS

Image encryption and decryption via aforesaid model improves the image transfer between two users. This is a very robust strategy for novel digital media like art, real estate papers, intellectual properties, and others. However, just a superimposition of a single image could be reverse engineered. Thus, this paper involves superimposition of two images prior to main image using different encryption algorithms, implying the benefits of the proposed framework.

Although detailed experimental evaluation is beyond the scope of this paper, peer-reviewed studies support our observations and the novelty of the proposed algorithm. Encryption times for alpha blending and our encryption-decryption method are approximately 0.29 sec and 0.45 sec, respectively, resulting in a total added processing time of around 1 second. However, this time is subject to vary based on image resolution and hardware capabilities of the system used. In comparison with state-of-the-art methods, the median Ethereum NFT transaction times are approximately 57 sec, with 90% of transactions completing within 8 minutes. Thus, the added latency represents only about 1.75% of the median transfer time, demonstrating that the proposed algorithm introduces a minimal overhead.

C. ROBUSTNESS ANALYSIS

It is imperative to perform a robustness analysis for image encryption in order to thoroughly assess the effectiveness and dependability of encryption methods, especially with regard to safe image transmission and storage. Robustness metrics like Peak Signal-to-Noise Ratio (PSNR) and Bit Error Rate (BER) play pivotal roles in this evaluation process. Such analyses are crucial for assessing the resilience of encrypted images against diverse forms of attacks, potential transmission errors, and other vulnerabilities those may compromise the security.

PSNR measures the degree of similarity between encrypted and original pictures, with higher values denoting greater fidelity. On the other hand, BER gauges transmission accuracy by calculating the ratio of bits that are sent wrongly to all bits that are sent. By using these measurements, researchers can identify potential vulnerabilities in encryption algorithms, conduct a methodical analysis of their robustness, and suggest enhancements to guarantee the secrecy and integrity of encrypted image data.

(a) **GAN's Major Weaknesses:** GANs are vulnerable to several types of attacks that can compromise their reliability.

Adversarial examples, where inputs are subtly modified to mislead the model, can disrupt the superimposition process.

(i) **Poisoning attacks** inject maliciously crafted training data to manipulate outputs and degrade model performance.

(ii) **Model inversion or reconstruction attacks** exploit information leaked during the GAN's operation to recover original input data, even if encryption is applied. If attackers succeed in recreating a GAN model, they can reconstruct sensitive training data, perform membership or property inference, or craft stronger adversarial examples. For instance, subtly altering inputs so the GAN produces incorrect or biased outputs while appearing normal. They may even repurpose the model for intellectual property theft or backdoor attacks. However, robustness of GANs can be enhanced through multiple complementary measures.

(iii) **Adversarial training** which incorporates crafted adversarial examples during training, improves model resilience without substantially affecting performance.

(iv) **Data sanitization and regular curation** of training datasets help prevent poisoning attacks by identifying and removing malicious samples before they impact the model.

(v) **Privacy-preserving** methods, such as differential privacy, limit information leakage and reduce the risk of model inversion or reconstruction attacks. Continuous monitoring and retraining detects anomalous patterns and emerging threats, maintaining long-term model reliability.

(b) **Pattern analysis:** If the Gaussian noise pattern is predictable or weak, attackers might analyze and reverse-engineer it. Use a strong and unpredictable noise pattern to resist attacks.

(c) **Attacks in Blockchain:** *Double-spending* attacks occur when an attacker spends the same NFT multiple times. Thus, it should be ensured that the blockchain technology offers protections against this attack.

On the other hand, *51 percent attack* could allow an attacker to control the blockchain and manipulate transactions. Thus, a user must aware of vulnerabilities in NFT platform used for a transfer. To address any security issues, continuous monitoring and updating the NFT platform is essential.

V. CONCLUSION

In this study, we have proposed an approach for securely transferring images through the integration of GANs, Gaussian noise addition, and NFT technology on a blockchain. Our method introduces an additional layer of security leveraging the capability of GANs to superimpose images onto random counterparts using Gaussian noise. Subsequently, the encrypted image is encapsulated as NFT for secure transmission. Through the integration of GANs, we have addressed the need for obfuscation and data confidentiality by generating visually indistinguishable superimposed images. The introduction of Gaussian noise adds an element of unpredictability, enhancing resistance against various attacks, including pattern analysis and reconstruction attempts.

The use of blockchain ensures traceability, prevents double-spending, and establishes a decentralized and tamper-resistant environment for image transactions. However, the proposed model bears some challenges and potential vulnerabilities. Consideration must be given to GAN vulnerabilities, noise pattern predictability, and robust key management to ensure the resilience of the encryption process. Additionally, continuous monitoring and updates are imperative to address emerging security threats and vulnerabilities in the blockchain and smart contract components. As the field of secure image transfer evolves, future research directions may explore enhancements to GAN architectures, noise addition strategies, and blockchain technologies to further fortify the model against sophisticated attacks.

In summary, our proposed model presents a promising avenue for secure images transfer, combining advancements in generative modeling, encryption techniques, and blockchain technology. While acknowledging the existing challenges, this research contributes to the ongoing exploration of innovative solutions for secure and privacy-preserving image transmission in emerging digital ecosystems.

ACKNOWLEDGMENT

This work was been supported by the Birla Institute of Technology and Science (BITS) Pilani, Pilani Campus, Rajasthan, 333 031, India, for providing Open Access facilities with necessary computational infrastructure.

REFERENCES

- [1] Q. Wang, R. Li, Q. Wang, and S. Chen, "Non-fungible token (NFT): Overview, evaluation, opportunities and challenges," 2021, *arXiv:2105.07447*.
- [2] J. Davies, H. Sharifi, A. Lyons, R. Forster, and O. K. S. M. Elsayed, "Non-fungible tokens: The missing ingredient for sustainable supply chains in the metaverse age?" *Transp. Res. E, Logistics Transp. Rev.*, vol. 182, Feb. 2024, Art. no. 103412.
- [3] B. S. Diba, J. D. Plabon, N. Tasnim, M. Hossain, D. Mistry, S. Sarker, M. F. Mridha, Y. Okuyama, and J. Shin, "From centralization to decentralization: Blockchain's role in transforming social media platforms," *IEEE Access*, vol. 13, pp. 80478–80507, 2025.
- [4] A. Ghosh, Lavanya, V. Hassija, V. Chamola, and A. El Saddik, "A survey on decentralized metaverse using blockchain and web 3.0 technologies, applications, and more," *IEEE Access*, vol. 12, pp. 146915–146948, 2024.
- [5] L. Ante, "The non-fungible token (NFT) market and its relationship with Bitcoin and Ethereum," *FinTech*, vol. 1, no. 3, pp. 216–224, Jun. 2022.
- [6] S. Nakamoto. (2018). *Bitcoin: A Peer-to-Peer Electronic Cash System*. Accessed: Jun. 6, 2024. [Online]. Available: <https://www.ussc.gov/sites/default/files/pdf/training/annual-national-training-seminar/2018/EmergingTechBitcoinCrypto.pdf>
- [7] D. V. Le and A. Gervais, "AMR: Autonomous coin mixer with privacy preserving reward distribution," in *Proc. 3rd ACM Conf. Adv. Financial Technol.*, Sep. 2021, pp. 142–155.
- [8] A. Battah, M. Madine, I. Yaqoob, K. Salah, H. R. Hasan, and R. Jayaraman, "Blockchain and NFTs for trusted ownership, trading, and access of AI models," *IEEE Access*, vol. 10, pp. 112230–112249, 2022.
- [9] U. Khalil, M. Uddin, O. A. Malik, and O. W. Hong, "A novel NFT solution for assets digitization and authentication in cyber-physical systems: Blueprint and evaluation," *IEEE Open J. Comput. Soc.*, vol. 5, pp. 131–143, 2024.
- [10] H. S. Galal and A. M. Youssef, "Aegis: Privacy-preserving market for non-fungible tokens," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 1, pp. 92–102, Jan. 2023.
- [11] E. Glover. (Sep. 2022). *NFT Art Theft: What Buyers and Artists Need to Know*. Accessed: Jun. 3, 2024. [Online]. Available: <https://builtin.com/articles/NFT-art-theft>
- [12] D. Z. Morris. (May 2023). *CoinDesk Turns 10: 2016—How the DAO Hack Changed Ethereum and Crypto*. Accessed: Jul. 3, 2024. [Online]. Available: <https://www.coindesk.com/consensus-magazine/2023/05/09/coindesk-turns-10-how-the-dao-hack-changed-ethereum-and-crypto/>
- [13] D. Das, P. Bose, N. Ruaro, C. Kruegel, and G. Vigna, "Understanding security issues in the NFT ecosystem," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Nov. 2022, pp. 667–681.
- [14] A. Bera, D. Bhattacharjee, and H. P. H. Shum, "Two-stage human verification using HandCAPTCHA and anti-spoofed finger biometrics with feature selection," *Expert Syst. Appl.*, vol. 171, Jun. 2021, Art. no. 114583.
- [15] P. Ray, A. Bera, D. Giri, and D. Bhattacharjee, "Style matching CAPTCHA: Match neural transferred styles to thwart intelligent attacks," *Multimedia Syst.*, vol. 29, no. 4, pp. 1865–1895, Aug. 2023.
- [16] R. K. Challa, "CAPTCHA: A systematic review," in *Proc. IEEE Int. Conf. Advent Trends Multidisciplinary Res. Innov. (ICATMRI)*, Dec. 2020, pp. 1–8.
- [17] S. Lee, J. Lim, and I. H. Suh, "Progressive feature matching: Incremental graph construction and optimization," *IEEE Trans. Image Process.*, vol. 29, pp. 6992–7005, 2020.
- [18] J. Bellagarda and A. M. Abu-Mahfouz, "Connect2NFT: A web-based, blockchain enabled NFT application with the aim of reducing fraud and ensuring authenticated social, non-human verified digital identity," *Mathematics*, vol. 10, no. 21, p. 3934, Oct. 2022.
- [19] S. Mathur, A. Mathur, and N. Agarwal, "A review of basic mathematical transformation used in image processing," *Int. J. Eng. Res. Technol.*, vol. 4, no. 12, pp. 1–4, 2018.
- [20] M. Salvi and K. Vaidyanathan, "Multi-layer alpha blending," in *Proc. 18th Meeting ACM SIGGRAPH Symp. Interact. 3D Graph. Games*, Mar. 2014, pp. 151–158.
- [21] A. You, J. K. Kim, I. H. Ryu, and T. K. Yoo, "Application of generative adversarial networks (GAN) for ophthalmology image domains: A survey," *Eye Vis.*, vol. 9, no. 1, pp. 1–19, Dec. 2022.
- [22] K. B. Sarachik, "The effect of Gaussian error in object recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 19, no. 4, pp. 289–301, Apr. 1997.
- [23] S. C. Jogu, A. Nishitha, and R. V. R. Chary. *Blockchain-Based Authentications Using MetaMask Wallet As a Tool*. Accessed: Jun. 3, 2024. [Online]. Available: <https://ijariie.com/AdminUploadPdf/BlockchainbasedAuthenticationsusingMetaMaskWalletasaToolijarie19383.pdf>
- [24] S. Bhujel and Y. Rahulamathavan, "A survey: Security, transparency, and scalability issues of NFT's and its marketplaces," *Sensors*, vol. 22, no. 22, p. 8833, Nov. 2022.
- [25] J. Williamson and K. Curran, "The role of multi-factor authentication for modern day security," *Semiconductor Sci. Inf. Devices*, vol. 3, no. 1, pp. 16–23, May 2021.
- [26] H. Taherdoost, "Non-fungible tokens (NFT): A systematic review," *Information*, vol. 14, no. 1, p. 26, Dec. 2022.
- [27] *Ethereum Whitepaper*. Accessed: Jul. 3, 2024. [Online]. Available: <https://ethereum.org/en/whitepaper/>
- [28] Ethereum Improvement Proposals. (Jan. 2018). *ERC-721: Non-Fungible Token Standard*. Accessed: Jul. 3, 2024. [Online]. Available: <https://eips.ethereum.org/EIPS/eip-721>
- [29] Ethereum Improvement Proposals. (Jun. 2018). *ERC-1155: Multi Token Standard*. Accessed: Jul. 3, 2023. [Online]. Available: <https://eips.ethereum.org/EIPS/eip-1155>
- [30] P. Gonserkewitz, E. Karger, and M. Jagals, "Non-fungible tokens: Use cases of NFTs and future research agenda," *Risk Governance Control: Financial Markets Institutions*, vol. 12, no. 3, pp. 8–18, 2022.
- [31] Y. Zhang, Z. Chen, L. Zhang, and X. Tong, "Visualizing non-fungible token ethics: A case study on CryptoPunks," *HCI Blockchain [Internet]*, Oct. 2022. [Online]. Available: <https://hci-blockchain.pubpub.org/pub/chinavis2022>
- [32] X.-J. Jiang and X. F. Liu, "CryptoKitties transaction network analysis: The rise and fall of the first blockchain game mania," *Frontiers Phys.*, vol. 9, Mar. 2021, Art. no. 631665.
- [33] *Buy, Sell & Trade Bitcoin & Other Crypto Currencies With Gemini's Platform*. Accessed: Jul. 3, 2024. [Online]. Available: <https://www.gemini.com/cryptopedia/nba-topshot-nft-flow-blockchain-nba-moments>

- [34] U. Khalil, O. A. Malik, O. W. Hong, and M. Uddin, "Leveraging a novel NFT-enabled blockchain architecture for the authentication of IoT assets in smart cities," *Sci. Rep.*, vol. 13, no. 1, p. 19785, Nov. 2023.
- [35] A. Musamih, K. Salah, R. Jayaraman, S. Ellaham, M. Omar, and I. Yaqoob, "Blockchain and NFT-based solution for genomic data management, sharing, and monetization," *IEEE Access*, vol. 13, pp. 35780–35804, 2025.
- [36] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial networks," *Commun. ACM*, vol. 63, no. 11, pp. 139–144, 2020.
- [37] S. Shahrar and K. Hayawi, "NFTGAN: Non-fungible token art generation using generative adversarial networks," in *Proc. 7th Int. Conf. Mach. Learn. Technol. (ICMLT)*, Mar. 2022, pp. 255–259.
- [38] S. Uttam, N. A. Goodman, M. A. Neifeld, C. Kim, R. John, J. Kim, and D. Brady, "Optically multiplexed imaging with superposition space tracking," *Opt. Exp.*, vol. 17, no. 3, p. 1691, 2009.
- [39] Y. Rachlin, V. Shah, R. Hamilton Shepard, and T. Shih, *Dynamic Optically Multiplexed Imaging*. Accessed: Jun. 6, 2024. [Online]. Available: <https://apps.dtic.mil/sti/pdfs/AD1034764.pdf>
- [40] A. N. Hirani and T. Totsuka, "Combining frequency and spatial domain information for fast interactive image noise removal," in *Proc. 23rd Annu. Conf. Comput. Graph. Interact. Techn.*, Aug. 1996, pp. 269–276.
- [41] S. Pergoloni, A. Petroni, T.-C. Bui, G. Scarano, R. Cusani, and M. Biagi, "ASK-based spatial multiplexing RGB scheme using symbol-dependent self-interference for detection," *Opt. Exp.*, vol. 25, no. 13, p. 15028, 2017.
- [42] P. Shete and S. Kohle, "Image encryption using AES algorithm: Study and evaluation," *Int. J. Res. Appl. Sci. Eng. Technol.*, vol. 10, no. 9, pp. 1134–1137, Sep. 2022.
- [43] D. Horita, K. Aizawa, R. Suzuki, T. Yonetsuji, and H. Zhu, *Fast Nonlinear Image Unblending*. Accessed: Jun. 6, 2024. [Online]. Available: <https://openaccess.thecvf.com/content/WACV2022/papers/HoritaFastNonlinearImageUnblendingWACV2022paper.pdf>
- [44] M. Muja and D. G. Lowe, "Scalable nearest neighbor algorithms for high dimensional data," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 11, pp. 2227–2240, Nov. 2014.
- [45] M. Muja and D. G. Lowe, *Fast Matching of Binary Features*. Accessed: Jul. 4, 2024. [Online]. Available: <https://www.cs.ubc.ca/research/flann/uploads/FLANN/binarymatchingcv2012.pdf>
- [46] V. Vijayan and P. Kp, "FLANN based matching with SIFT descriptors for drowsy features extraction," in *Proc. 5th Int. Conf. Image Inf. Process. (ICIIP)*, Nov. 2019, pp. 600–605.
- [47] J. Luo, and O. Gwun, "A comparison of sift, pca-sift and surf," *Int. J. Image Processing*, vol. 3, no. 4, pp. 143–152, 2009.
- [48] S. Wang, Z. Guo, and Y. Liu, "An image matching method based on SIFT feature extraction and FLANN search algorithm improvement," *J. Phys., Conf. Ser.*, vol. 2037, no. 1, Sep. 2021, Art. no. 012122.
- [49] J. Koehler and R. Hauser, "Untangling unstructured cyclic flows—A solution based on continuations," in *On the Move To Meaningful Internet Systems 2004: CoopIS, DOA, and ODBASE*. Berlin, Germany: Springer, 2004, pp. 121–138.
- [50] D. Kiela and S. Clark, *Multi- and Cross-modal Semantics Beyond Vision: Grounding in Auditory Perception*. Accessed: Jul. 4, 2024. [Online]. Available: <https://aclanthology.org/D15-1293.pdf>
- [51] G. P. Renieblas, A. T. Nogués, A. M. González, N. Gómez-Leon, and E. G. del Castillo, "Structural similarity index family for image quality assessment in radiological images," *J. Med. Imag.*, vol. 4, no. 3, Jul. 2017, Art. no. 035501.
- [52] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.
- [53] A. Bera, R. Dey, D. Bhattacharjee, M. Nasipuri, and H. P. H. Shum, "Spoofing detection on hand images using quality assessment," *Multimedia Tools Appl.*, vol. 80, no. 19, pp. 28603–28626, Aug. 2021.
- [54] S. M. Smith and J. M. Brady, "SUSAN—A new approach to low level image processing," *Int. J. Comput. Vis.*, vol. 23, no. 1, pp. 45–78, May 1997.
- [55] Z. Wang, E. P. Simoncelli, and A. C. Bovik, "Multiscale structural similarity for image quality assessment," in *Proc. 37th Asilomar Conf. Signals, Syst. Comput.*, 2003, pp. 1398–1402.
- [56] M. Joshi, C. Shakher, and K. Singh, "Fractional Fourier transform based image multiplexing and encryption technique for four-color images using input images as keys," *Opt. Commun.*, vol. 283, no. 12, pp. 2496–2505, Jun. 2010.
- [57] A. H. Brahim, A. A. Pacha, and N. H. Said, "An image encryption scheme based on a modified AES algorithm by using a variable S-box," *J. Opt.*, vol. 53, no. 2, pp. 1170–1185, Apr. 2024.



ARNAV GUJARATHI received the degree in computer science from the Birla Institute of Technology and Science, Pilani, Pilani Campus, Rajasthan, India, in 2024. He is currently a Software Development Engineer with Amazon. Previously, he has interned with organizations, such as DRDO and several startups, where he explored cutting-edge technologies across domains, such as machine learning, computer vision, encryption, and front-end development. He enjoys solving real-world problems through code and have a growing interest in FinTech and sustainable tech applications. With a strong foundation in programming and a collaborative mindset, he worked on fast-paced, interdisciplinary projects.



PRATHAM OZA received the degree in computer science from the Birla Institute of Technology and Science, Pilani, Pilani Campus, Rajasthan, India, in 2024. Currently, he is the Head of Go-To-Market (GTM) with Fibr, an AI-powered Conversion Rate Optimization (CRO) Platform. During his graduation, he founded a blockchain-based startup that successfully secured equity-free grants totaling INR 3.5 lakhs. Since then, he has been maintaining a strong interest in emerging technologies, particularly in the areas of blockchain and artificial intelligence, and continue to explore their applications in business and product innovation.



ASISH BERA (Senior Member, IEEE) received the Ph.D. degree in computer science and engineering from Jadavpur University, Kolkata, India, in 2018. He was a Postdoctoral Research Associate with the Department of Computer Science, Edge Hill University, U.K. He is currently an Assistant Professor with the Department of Computer Science and Information Systems, Birla Institute of Technology and Science, Pilani, Pilani Campus, India. He has authored several articles in IEEE TRANSACTIONS and other reputed venues. As a Principal Investigator, he has received the Prime Minister Early Career Research Grant (PMECRG-2024) funded by the Anusandhan National Research Foundation (ANRF), Government of India. His current research interests include computer vision, generative AI, vision-language models (VLMs), and deep learning applications for agriculture, human activity recognition, and healthcare. As an Editorial Board Member of *Scientific Reports* and *Nature Portfolio*. He has received the Springer Nature Editor of Distinction Award 2025 for Author Services.

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